

NATIONAL UNIVERSITY OF SINGAPORE
SCHOOL OF COMPUTING
CS2106: INTRODUCTION TO OPERATING SYSTEMS
MIDTERM PRACTICE VARIANT A

Time Allowed: 1 Hour
Answer all questions. Questions 1-8 are MCQ; questions 9-12 are multiple-response True/False; questions 13-14 are short-answer questions.

MCQ 1 and 2 are based on this code:

```
int main() {  
    int a = fork();  
    printf("A %d\n", a);  
    int b = fork();  
    printf("B %d\n", b);  
    return 0;  
}
```

1. How many lines are printed in total?

- a. 2
- b. 3
- c. 4
- d. 6
- e. 8

2. Which statement is always true?

- a. Exactly two processes execute the second fork.
- b. The child of the first fork prints a non-zero value for a.
- c. Every process prints exactly one line.
- d. The two B lines have identical values.
- e. The program creates exactly three processes.

3. A running process executes a blocking read on an empty pipe. Which transition occurs?

- a. Running to Ready
- b. Running to Blocked
- c. Ready to Running
- d. Running to Terminated
- e. New to Ready

4. A process performs getpid(), assuming the call does not block. What is its process-state transition?

- a. Running to Blocked
- b. Running to Ready
- c. Running remains Running
- d. Running to New
- e. Ready to Terminated

5. Which situation can produce priority inversion?

- a. A high-priority task waits for a resource held by a low-priority task while a medium-priority task runs.
- b. Two high-priority tasks use different resources.
- c. A low-priority task waits for a resource held by a high-priority task.
- d. A medium-priority task terminates.
- e. A task performs only computation.

6. Which is a typical microkernel disadvantage?

- a. All services must run in kernel mode.
- b. User-space services cannot communicate.
- c. Extra IPC and context-switch overhead.
- d. Device drivers cannot be isolated.
- e. It cannot implement scheduling.

7. A parent runs 240 ms of CPU work before fork(). The child runs 80 ms afterward and the parent runs 120 ms afterward. Under non-preemptive FCFS, with fork itself CPU-only, which process finishes first?

- a. Child
- b. Parent

- c. They finish together
- d. Whichever has the larger PID
- e. Cannot be determined

8. Which statement about a semaphore initialized to 3 is correct?

- a. At most three processes can ever wait on it.
- b. At most three processes can be inside the protected section at once.
- c. Its value must become negative when a process waits.
- d. `signal()` decreases its value.
- e. `wait()` always blocks.

9. Mark each statement True or False about context-switch overhead:

- a. More saved CPU registers can increase the overhead.
- b. The number of user-level threads matters to a purely user-level thread switch performed without kernel involvement.
- c. A larger stack automatically means all stack contents are copied during every switch.
- d. Saving a process identifier takes time, but the number of children does not matter if only one parent PID is stored.

10. Mark each statement True or False about IPC:

- a. Shared memory usually requires explicit synchronization for shared writes.
- b. A pipe created by `pipe()` is normally unidirectional.
- c. Signals are suitable for transferring a large file.
- d. Message passing can require kernel involvement for each send/receive operation.

11. Semaphore S starts at 2. Every process executes `wait(S)`; section A; `signal(S)`.

- a. More than two processes may be blocked on S.
- b. At most two processes execute section A simultaneously.
- c. Section A must be a critical section.
- d. S can grow to 3 after repeated executions.

12. Mark each statement True or False:

- a. A system call may switch the CPU into kernel mode.
- b. A normal library function must always enter kernel mode.
- c. An empty program may still invoke exit-related OS code.
- d. A process can be ready without currently using the CPU.

13. Non-preemptive SJF uses exponential prediction: $\text{New} = \alpha \text{ Actual} + (1-\alpha) \text{ Old}$. Processes A, B, C have predictions 160 ms, 120 ms, and 90 ms. B was blocked and just used 200 ms of CPU.

- a. Find B's new prediction for $\alpha=0.25$.
- b. Which process is selected next?
- c. When should actual CPU usage be recorded?
- d. If workload phases change rapidly, should α be larger or smaller? Explain briefly.

14. x starts at 50. Two processes execute concurrently. Determine all possible final values of x, and the minimum and maximum, for each case:

(a) Both processes use the same binary semaphore initialized to 1:

`wait(S); x=x+5; x=x-2; signal(S);`

(b) Process A uses S1 and process B uses S2; both start at 1, and each performs the same two updates without sharing a semaphore. Explain why the answer differs.